

Code

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```
# read in packages
library(tidyverse)
library(here)
library(janitor)
library(lubridate)
library(showtext)
library(treemap)
library(viridis)
library(ggthemes)

# read in my data and store it as an object called ben-data
ben_data <- read_csv(here("data", "ben-data-tracker - Sheet1.csv"))
```

Data Cleaning

Goals:

Clean column names

Convert dates into an ordered variable

Convert day of week to ordered variable

```
# create new object called ben_clean from ben_data
ben_clean <- ben_data |>
  # remove capital letters, add underscores
  clean_names() |>
  # convert dates into an ordered variable
  mutate(date_clean = ymd(date)) |>
  # set day of the week to have levels by chronological order
  mutate(day_of_week = as_factor(day_of_week),
         day_of_week = fct_relevel(day_of_week,
                                   "Monday",
                                   "Tuesday",
                                   "Wednesday",
                                   "Thursday",
                                   "Friday",
                                   "Saturday",
                                   "Sunday")) |>

  # remove unneeded columns
  subset(select = -date)

# check data frame's structure to ensure columns are read correctly
str(ben_clean)
```

```
'data.frame':  32 obs. of  7 variables:
 $ distance_walked_km: num  8.03 6.23 4.49 4.29 7.04 ...
 $ day_of_week       : Factor w/ 7 levels "Monday","Tuesday",...: 4 5 6 7 1 2 3 4 5 6 ...
 $ bus_trips         : int   3 0 0 0 0 2 2 3 2 0 ...
 $ calendar_events   : int   5 4 1 0 3 6 4 4 4 1 ...
 $ high_temperature_c: int   22 21 16 18 19 21 21 22 19 18 ...
 $ lunch_location    : chr   "IV" "Campus" "IV" "Campus" ...
 $ date_clean        : Date, format: "2026-04-23" "2026-04-24" ...
```

```
# add custom fonts
font_add_google("Darker Grotesque", "grot")
font_add_google("Lexend", "lex")
# use showtext to render fonts
showtext_auto()
```

Day of week and date now have associated number values, so the data can be ordered. Each figure will use different columns, so I will begin each plot with a select function.

Figure 1 - Timeseries

Variables being used: distance walked, date, day of week

```
# create new object to only include needed columns
ben_time <- ben_clean |>
  # select needed columns
  select(date_clean, distance_walked_km, day_of_week)
```

```
# add base layer using cleaned data for this chart
# x-axis: date
# y-axis: distance walked
# color: by day of week (for points)
ggplot(data = ben_time,
  mapping = aes(
    x = date_clean,
    y = distance_walked_km,
    color = day_of_week
  )) +
  # add the distance walked information
  geom_line(
    # use the distance walked column
    mapping = aes(y = distance_walked_km),
    # grey colored line
    color = "grey10",
    # adjust the line's width
    linewidth = 0.3
  ) +
  # add points
  geom_point() +
  # color points in rainbow order by day of week
  scale_color_manual(values = c(
    "Monday" = "red",
    "Tuesday" = "orange2",
    "Wednesday" = "yellow",
    "Thursday" = "green3",
    "Friday" = "cyan4",
    "Saturday" = "blue1",
    "Sunday" = "purple3"
  )) +
  # adjust y-axis to start at 0
  ylim(0, 14) +
```

```

# adjust color of the background and panel grid to be the same
# hides grid
theme(
  panel.background = element_rect(fill = "#D1BCA1"),
  panel.grid = element_line(color = "#D1BCA1")) +
# creating title
# renaming axes
# renaming legend (mapped to color)
labs(
  x = "Date",
  y = "Distance Walked (km)",
  color = "Day Of Week",
  title = "Daily Distance Walked"
) +
# add an annotation
annotate(
  # annotation uses curve geometry
  geom = "curve",
  # sets the beginning x location of the curve (must use date format to
  # align with chart)
  x = as.Date("2026-05-08"),
  # sets the beginning y location of the curve
  y = 11.25,
  # sets the end x location of the curve
  xend = as.Date("2026-05-15"),
  # sets the end y location of the curve
  yend = 13.13,
  # sets direction of curve
  curvature = -0.5,
  # adjusts arrow length
  arrow = arrow(length = unit(4, "mm"))) +
# add text annotation
annotate(
  # use text geometry
  geom = "text",
  # x location of text
  x = as.Date("2026-05-08"),
  # y location of text
  y = 11,
  # actual text
  label = "Hiking Day",
  # center text around set x location

```

```

hjust = "center",
# use larger size
size = 8,
# use custom font
family = "grot") +
# annotation 2, same format as before, so fewer annotations
# curve and arrow geometry
annotate(
  geom = "curve",
  x = as.Date("2026-05-17"),
  y = 2.5,
  xend = as.Date("2026-05-11"),
  yend = 3.33,
  curvature = 0.5,
  arrow = arrow(length = unit(4, "mm"))) +
annotate(
  geom = "text",
  x = as.Date("2026-05-17"),
  y = 1,
  # actual text with line break
  label = "Less walking \non Sundays (usually)",
  hjust = "center",
  size = 5,
  family = "grot")

```

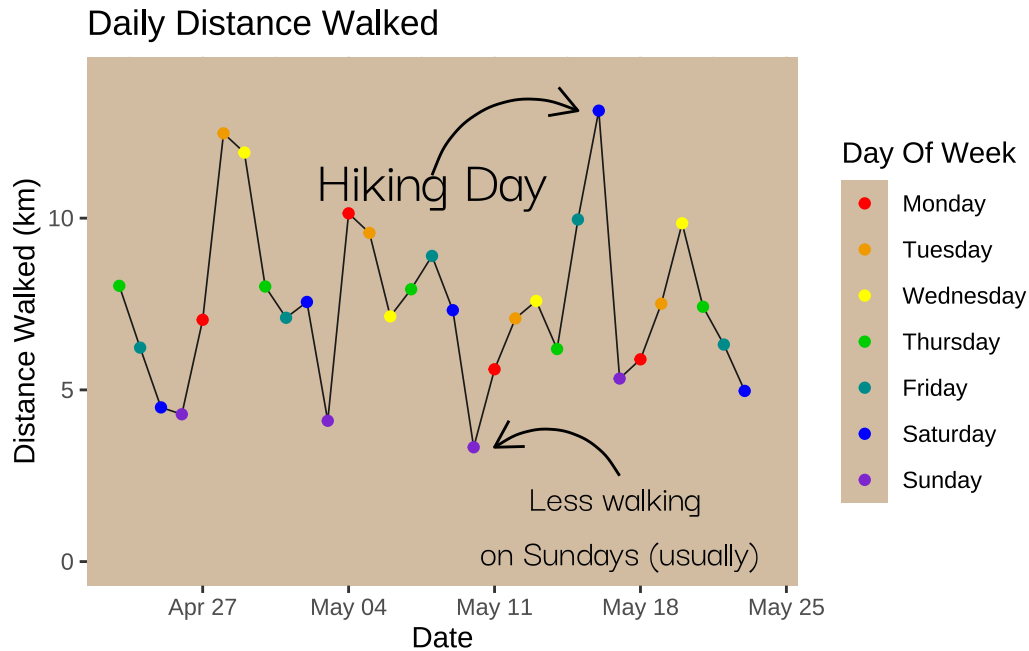


Figure 2 - Treemap

Variables being used: distance walked, bus trips, lunch location

```
# create new object to only include needed columns
ben_tree <- ben_clean |>
  # select needed columns
  select(distance_walked_km, bus_trips, lunch_location)
```

```
# create a treemap diagram
# use ben_tree for the data
treemap(ben_tree,
  # hierarchy of variables to display
  index = c("lunch_location", "bus_trips", "distance_walked_km"),
  # box size controlled by distance walked
  vSize = "distance_walked_km",
  # creates plot title
  title = "Distance Walked (km) by Lunch Location (black) and Bus Trips (white)",
  fontsize.title = 12,
  # sets the color palette based on layer 1 (lunch location)
  palette = c("#2F2ABF", "#2C6B36", "#C21D1D", "#BFAB1B"),
```

```

# sets border colors for the 3 layers
# black for first 2, grey for 3rd
border.col=c("black", "black", "grey50"),
# adjust border widths, higher layers are thicker
border.lwds=c(4,2.5,1),
# adjusts font size on labels, 3rd layer has no labels since size already
# accounts for the value
fontsize.labels=c(20, 15, 0),
# sets layer 1 text to black and layer 2 to white
fontcolor.labels=c("black", "white"),
# removes highlighting on labels
bg.labels=c("transparent"),
# layer 1's labels are centered, layer 2 in top left of box
align.labels=list( c("center", "center"), c("left", "top")),
fontfamily.labels = "lex",
)

```

Distance Walked (km) by Lunch Location (black) and Bus Trips (white)

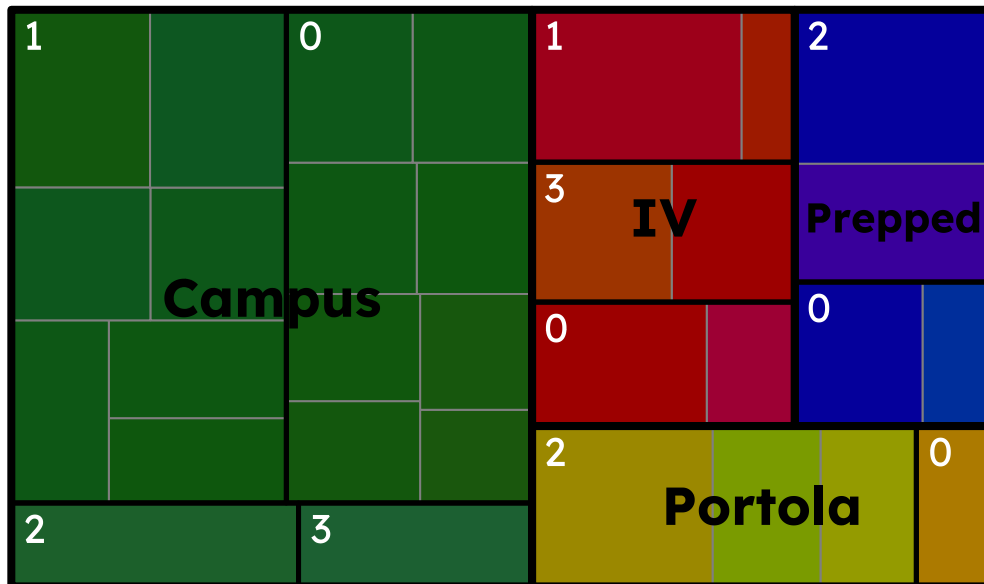


Figure 3 - Scatterplot

Variables used: distance_walked, day_of_week, calendar_events

```

# create new object to only include needed columns
ben_scatter <- ben_clean |>
  # select needed columns
  select(distance_walked_km, day_of_week, calendar_events)

# create a base layer for a scatterplot
# data: use ben_scatter
# x-axis: calendar events
# y-axis: distance walked
# color: group by day of week
ggplot(
  data = ben_scatter,
  mapping = aes(
    x = calendar_events,
    y = distance_walked_km,
    color = day_of_week
  )) +
  # create scatterplot
  geom_point() +
  # adjust colors, same as figure 1
  scale_color_manual(values = c(
    "Monday" = "red",
    "Tuesday" = "orange2",
    "Wednesday" = "yellow",
    "Thursday" = "green3",
    "Friday" = "cyan4",
    "Saturday" = "blue1",
    "Sunday" = "purple3"
  )) +
  # adjust background color and remove panel gridlines
  theme(
    panel.background = element_rect(fill = "#DEBADB"),
    panel.grid = element_line(color = "#DEBADB")) +
  # rename axes and legend, create title
  labs(
    x = "Number of Google Calendar Events",
    y = "Distance Walked (km)",
    title = "Distance Walked and Calendar Events by Day of Week",
    color = "Day of Week"
  ) +
  # adjust y-axis limits so it starts at 0
  ylim(0, 14) +

```



```

# annotate with a vertical line to highlight weekday vs weekend pattern
# xintercept sets the location
# color sets the color to black
# linetype makes it dotted
geom_vline(
  xintercept = 2.5,
  color = "black",
  linetype = "dotted"
) +
# since I annotated these in detail in figure 1, I'll make them sparse here
# creating the weekend annotation
annotate(
  geom = "text",
  x = 1,
  y = 2.5,
  label = "Weekends",
  hjust = "center",
  size = 8,
  family = "grot") +
# create annotation for weekday label
annotate(
  geom = "text",
  x = 5,
  y = 2.5,
  label = "Weekdays",
  hjust = "center",
  size = 8,
  family = "grot") +
# create annotation describing the pattern
annotate(
  geom = "text",
  x = 5,
  y = 0.5,
  # line break needed
  label = "weekday calendar events are \nstrictly higher than weekends",
  hjust = "center",
  size = 4,
  family = "grot")

```

Distance Walked and Calendar Events by Day of Week

